Math 372: Homework 09

Due Friday, October 23

Problem 1 (Waiting times)**.** Let T be the time, in minutes, between successive drink orders at a bar. Suppose that a reasonable probability model for T is given by the pdf

$$f_T(t) = \begin{cases} Ce^{-\frac{t}{4}} & : \quad t \geq 0 \\ 0 & : \quad t < 0, \end{cases}$$

where is $C > 0$ is some number that you'll have to determine.

(a) Find a value for C such that $f_T(t)$ is a valid pdf.

(b) What is the probability that the time between successive orders is greater than 4 minutes?

(c) On average, how long must the bartender wait between drink orders?

Problem 2. Let X denote the amount of time a book on two-hour reserve is actually checked out, and suppose the cdf of X is

$$F(x) = \begin{cases} 0 & : \quad x < 0 \\ \frac{x^2}{4} & : \quad 0 \leq x < 2 \\ 1 & : \quad 2 \leq x \end{cases}$$

(a) Find $\mathbb{P}[X \leq 1]$

(b) Find $\mathbb{P}[.5 \leq X \leq 1]$

(c) What is the median checkout duration of $\tilde{\mu}$? [In other words, solve $F(\tilde{\mu}) = .5$.]

(d) Find the density function $f(x)$

(e) Calculate $\mathbb{E}[X]$

(f) Calculate $\text{Var}(X)$.

(g) If the borrower is charged an an amount $h(X) = X^2$ when checkout duration is X, compute the expected charge $\mathbb{E}[h(X)]$.

Problem 3. Let T be the waiting time until the lightbulb spontaneoulsy burns out. Assume that T is exponentially distributed and that the expected lifetime of the bulb is 2 years.

(a) What is the rate parameter of T?

(b) What is the probability density function of T? What is the cumulative density function?

(c) Compute $\mathbb{P}[1 < T < 2]$.

(d) Let σ denote the standard deviation of T. Compute σ.

(e) Compute $\mathbb{P}[T \leq 4\sigma]$

(f) Compute $\mathbb{P}[T > 2]$.

(g) Compute $\mathbb{P}[T > 4 \mid T > 2]$.

(h) Using several complete sentences, interpret your answers to parts (f) and (g). Is there a contradiction?

Problem 4 (Exercise 4.1.2)**.** Suppose the reaction temperature X (in celsius) in a certain chemical process has uniform distribution with $A = -5$ and $B = 5$.

(a) Compute $\mathbb{P}[X < 0]$

(b) Compute $\mathbb{P}[-2.5 < X < 2.5]$

(c) Compute $\mathbb{P}[-2 \leq X \leq 3]$

(d) For k satisfying $-5 < k < k + 4 < 5$, compute $\mathbb{P}[k < X < k + 4]$.

Problem 5. The lifetime of a gadget costing \$300 is exponentially distributed with mean 3 years. The manufacturer agrees to pay a full refund to a buyer if the gadget fails during the first year following the purchase, a two-thirds refund if it fails between the first and second year, and a one-third refund if it fails between the second and third year. If the manufacturer sells 1000 gadgets, how much should it expect to pay in refunds?